



Penrith Selective High
School

2018

Higher School Certificate

Trial Examination

Mathematics Extension 1

General instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen.
- NESA approved calculators may be used.
- A reference sheet is provided with this examination paper.
- In questions 11-14, show relevant mathematical reasoning and/or calculations.
- **No** correction tape to be used.

Total marks:

70

Section I – 10 marks (pages 1 - 4)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 5 - 8)

- Attempt Questions 11-14
- Allow about 1 hour and 45 minutes for this section

	Preliminary	Polynomials	Binomial Theorem	Inverse Trigonometry	Calculus	Probability	Mathematical Induction	
MC Q1-10	Q1,3 4,5 /4			Q6, 7, 9 /3	Q2,10 /2	Q8 /1		
Question 11	a /3			f /3	b,c,d /8	e /1		
Question 12	a,b /6		d /3	c /6				
Question 13	d /3	b /5		a, c /7				
Question 14					a,b /8	c /4	d /3	170
Total	/16	/5	/3	/19	/18	/6	/3	%

Student Number: _____

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

- 1 The point P divides the interval from $A(-7, 6)$ and $B(4, -6)$ **externally** in the ratio 2: 3.

What is the x -coordinate of P ?

- (A) -29
- (B) 15
- (C) 20
- (D) 22

- 2 What is the value of $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$?

- (A) 0
- (B) $\frac{3}{5}$
- (C) 1
- (D) $\frac{5}{3}$

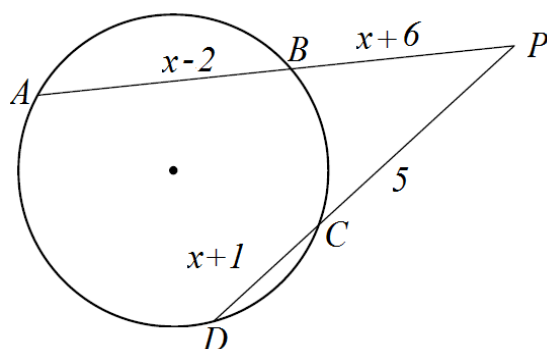
- 3 The Cartesian equation of the tangent, at $t = -2$, to the parabola $x = t - 4$, $y = t^2 + 5$ is:

- (A) $4x - y + 15 = 0$
- (B) $4x + y - 15 = 0$
- (C) $4x - y - 15 = 0$
- (D) $4x + y + 15 = 0$

4 What are the asymptotes of $y = \frac{5x}{(x+7)(3x-1)}$?

- (A) $y = 0, x = -7, x = \frac{1}{3}$
- (B) $y = 0, x = 7, x = -\frac{1}{3}$
- (C) $y = 5, x = -7, x = -\frac{1}{3}$
- (D) $y = 5, x = -7, x = \frac{1}{3}$

5 Two secants from an external point P cut off intervals on a circle as shown below.



What is the value of x ?

- (A) $\frac{1}{2}$
- (B) $\frac{1 + \sqrt{69}}{2}$
- (C) $\frac{3}{2}$
- (D) 6

6 Which of the following is the derivative of $\tan^{-1}(e^{-2x})$?

- (A) $\frac{e^{2x}}{1 + e^{2x}}$
- (B) $\frac{-e^{-2x}}{1 + e^{-2x}}$
- (C) $\frac{-2e^{-2x}}{1 + e^{4x}}$
- (D) $\frac{-2e^{-2x}}{1 + e^{-4x}}$

- 7 Let $|b| \leq 1$. What is the general solution to $\cos \frac{x}{3} = b$?
- (A) $k\pi \pm \cos^{-1} b$
- (B) $2k\pi \pm 3 \cos^{-1} b$
- (C) $4k\pi \pm \cos^{-1} b$
- (D) $6k\pi \pm 3 \cos^{-1} b$
- 8 Adrian, Bryson and six friends arrange themselves at random in a circle. What is the probability that Adrian and Bryson are *not* together?
- (A) $\frac{1}{5040}$
- (B) $\frac{5}{7}$
- (C) $\frac{2}{7}$
- (D) $\frac{5039}{5040}$
- 9 The primitive function of $\frac{1}{x^2 - 6x + 12}$ is:
- (A) $\ln(x^2 - 6x + 12) + C$
- (B) $\frac{1}{3} \tan^{-1}\left(\frac{x-3}{3}\right) + C$
- (C) $\frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{x-3}{\sqrt{3}}\right) + C$
- (D) $\frac{1}{9} \tan^{-1}\left(\frac{x-3}{9}\right) + C$
- 10 Given that $\frac{g'(x)}{g(x)} = 2$, which of the following statements are true?
(Note: C is a constant in each case)
- (A) $g(x) = e^x + C$
- (B) $g(x) = e^{2x} \times C$
- (C) $g(x) = 2 \ln x + C$
- (D) $g(x) = C \ln x$

End of Section I

Section II

60 marks

Attempt Questions 11-14

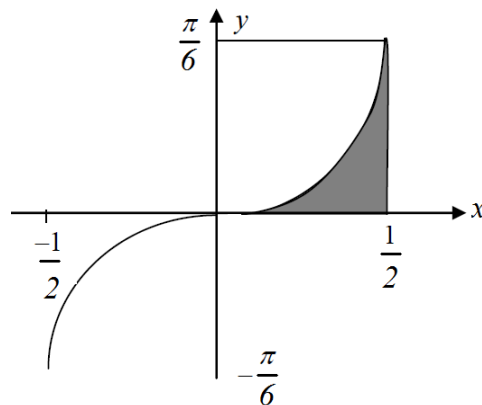
Allow about 1 hour and 45 minutes for this section

Answer each question on a **SEPARATE** booklet.

In Questions 11-14, your responses should include relevant mathematical reasonings and/or calculations.

Question 11 (15 marks)

- (a) Solve the inequality: $\frac{5x}{x-2} \leq 4$ 3
- (b) Find $\int \cos^2 6x \, dx$ 2
- (c) Evaluate $\int_0^2 \frac{3x}{(3x+1)^2} \, dx$ by using the substitution $u = 3x + 1$. 3
- (d) A circular oil slick lies on the surface of a body of water. Its area is increasing at the rate of $16 \, \text{m}^2/\text{min}$. At what rate is the radius increasing at the time the radius is 5 metres to 2 decimal places ? 3
- (e) Colour blindness affects 8% of all men. What is the expression of the probability that any random sample of 14 males should contain exactly 6 males that are colour blind? 1
- (f) The shaded area shown in the diagram below is bounded by the x -axis, $y = \sin^{-1} x$ and the line $x = \frac{1}{2}$. 1



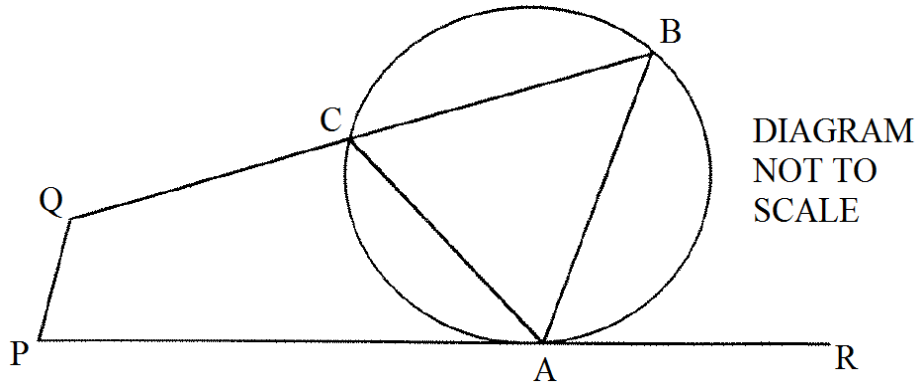
Find the exact value of the shaded area.

3

End of Question 11

Question 12 (15 marks)

- (a) ABC is a triangle inscribed in a circle. PA is a tangent to the circle. PQ is drawn parallel to AB and meets BC produced to Q .
Copy the diagram into your booklet.



Prove $APQC$ is a cyclic quadrilateral.

2

- (b) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $4ay = x^2$ such that the tangents at P and Q intersect at an angle of 45° . Let T be the point of intersection. The tangent at P is $y = px - ap^2$. (DO NOT PROVE THIS).

i) Show that $p - q = 1 + pq$

2

ii) Find the locus of T .

2

- (c) i) Sketch the curve $y = \cos^{-1}\left(\frac{x-2}{2}\right)$

2

ii) Show that when this curve is rotated about the y -axis, the volume of solid of revolution generated is $6\pi^2$ cubic units.

4

- (d) Show that

3

$$\binom{n}{0} + 2\binom{n}{1} + 3\binom{n}{2} + \dots + (n+1)\binom{n}{n} = (n+2)2^{n-1}$$

End of Question 12

Question 13 (15 marks)

(a) Find the exact value of $\sin\left(\cos^{-1}\frac{2}{3} + \tan^{-1}\left(-\frac{3}{4}\right)\right)$ 2

(b) A monic cubic polynomial when divided by $x^2 + 7$ leaves a remainder of $x + 12$ and when divided by x leaves a remainder of -6 .

i) Find the polynomial in the form $ax^3 + bx^2 + cx + d$. 3

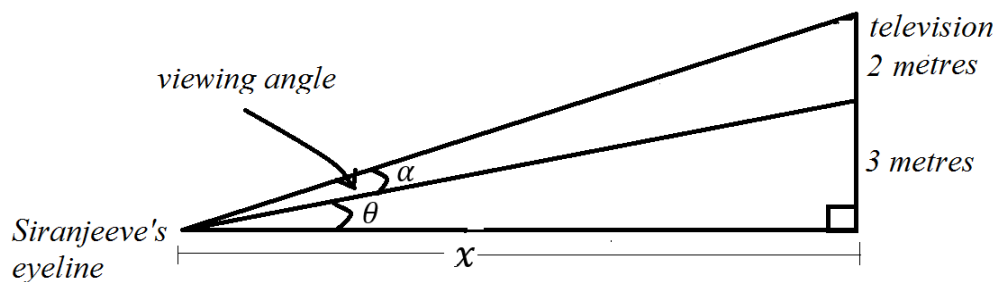
ii) The polynomial above has a root close to $x = 1$. Using one application of Newton's Method, give a better approximation to 2 decimal places. 2

(c) Siranjeeve buys a 2 metre tall LCD television screen for his cinema room. He mounts it on a vertical wall, placing it so that the base of the television is 3 metres above his horizontal eye line from where he sits in his favourite armchair. Let the distance from his eye to the wall be x metres and the angle from his eye to the top and base of the television be α (the viewing angle).

Let θ be the angle of elevation to the base of the television.

i) Show that the angle of vision α is given by 1

$$\alpha = \tan^{-1}\left(\frac{5}{x}\right) - \tan^{-1}\left(\frac{3}{x}\right)$$



ii) Show that for a maximum viewing angle α , $x = \sqrt{15}$ metres. 3

iii) Hence, find the maximum viewing angle α , to the nearest degree. 1

(d) Solve for x : $\log_{\frac{1}{2}}\left(\frac{1}{x}\right) \geq \log_2(4x - 1)$ 3

End of Question 13

Question 14 (15 marks)

- (a) A certain particle moves along the x -axis according to the equation $t = 4x^2 - 6x + 3$ where x is measured in centimetres and t in seconds. Initially the particle is 1.5 cm to the right of the origin O and moving away from O .

i) Prove that the velocity, $v\text{ cms}^{-1}$ is given by $v = \frac{1}{8x-6}$. 1

ii) Find an expression for the acceleration, $a\text{ cms}^{-2}$, in terms of x 2

iii) Find the velocity of the particle when $t = 8$ seconds. 2

- (b) i) Show that the function $T = R + Ae^{-kt}$ is a solution to the differential equation 1

$$\frac{dT}{dt} = -k(T - R)$$

ii) A metal cake tin is removed from an oven at a temperature of 190°C . If the cake tin takes five minutes to cool to 85°C and the room temperature is 25°C , find the time (to the nearest minute) it takes for the cake tin to cool to 63°C . 2

(Assume that the cake tin cools at a rate proportional to the difference between the temperature of the cake tin and the temperature of the surrounding air.)

- (c) A die is biased so that in any single throw the probability of an odd number is p where p is a constant such that $0 < p < 1$, $p \neq 0.5$.

i) Show that in six throws of the die the probability of at most one even number is $6p^5 - 5p^6$. 2

ii) Find the probability that in six throws of the die the product of the scores is even. 2

- (d) Prove by Mathematical Induction

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} \leq 2 - \frac{1}{n}, \text{ for all integers } n \geq 1. \quad \text{3}$$

End of Question 14

End of Examination

Find the exact value of the volume of the solid of revolution formed when the region bounded by the curve $y = 3 \sec \frac{x}{2}$, the x -axis and the lines $x = 0$ and $x = \pi$ is rotated about the x -axis. 3

Find the value of k such that $\tan^{-1}(k) + \tan^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{4}$ 3

In the expansion of $\left(x^3 + \frac{1}{x}\right)^7$, does the expression contain a constant term? Justify your answer. 2

A particle is moving in a straight line. At time t seconds it has displacement x metres where from a fixed point O on the line, velocity $v \text{ ms}^{-1}$ given by $v = \cos^2 x$ and acceleration $a \text{ ms}^{-2}$. The particle starts at O .

i) Find expressions for a in terms of x for x in terms of t 2

ii) Sketch the graph of x against t . 1

ii) Describe the motion of the particle from its initial position to its limiting position. 2

i) Show that the function $T = R + Ae^{-kt}$ is a solution to the differential equation 1

$$\frac{dT}{dt} = -k(T - R)$$

ii) A metal cake tin is removed from an oven at a temperature of 190°C . If the cake tin takes ten minutes to cool to 155°C and the room temperature is 25°C , find the time (to the nearest minute) it takes for the cake tin to cool to 90°C . 2

(Assume that the cake tin cools at a rate proportional to the difference between the temperature of the cake tin and the temperature of the surrounding air.)

Examination: TRIAL HSC.

Level: Extension One

Year: 2018

M.C.

1A 2B 3D 4A 5A

6D 7D 8B 9C 10B

 $\frac{1}{2}$

QUESTION: 11

Markers Comments

$$(a) \frac{5x}{x-2} \leq 4 \quad x \neq 2$$

$$5x(x-2) \leq 4(x-2)^2$$

$$5x(x-2) - 4(x-2)^2 \leq 0 \quad (1)$$

$$(x-2)[5x - 4(x-2)] \leq 0$$

$$(x-2)(x+8) \leq 0 \quad (1)$$



$$-8 \leq x < 2 \quad (1)$$

* If students wrote $-8 \leq x \leq 2$ they scored 2 marks.

$$(b) \int \cos^2 6x \, dx = \frac{1}{2} \int (1 + \cos 12x) \, dx \quad (1)$$

$$= \frac{1}{2} (x + \frac{1}{12} \sin 12x) + C$$

$$= \frac{1}{2}x + \frac{1}{24} \sin 12x + C \quad (1)$$

$$(c) u = 3x + 1$$

$$\text{when } x = 2, u = 7$$

$$\frac{du}{dx} = 3$$

$$\text{when } x = 0, u = 1 \quad (1)$$

$$du = 3dx$$

$$\int_0^2 \frac{3x}{(3x+1)^2} \, dx = \frac{1}{3} \int_1^7 \frac{u-1}{u^2} \, du$$

$$= \frac{1}{3} \int_1^7 \left(\frac{1}{u} - \frac{1}{u^2} \right) \, du$$

$$= \frac{1}{3} \int_1^7 \left(\frac{1}{u} - u^{-2} \right) \, du$$

$$= \frac{1}{3} [\ln u + u^{-1}]_1^7 \quad (1)$$

$$= \frac{1}{3} \left[\ln 7 + \frac{1}{7} - (\ln 1 + 1) \right]$$

$$= \frac{1}{3} \left(\ln 7 + \frac{6}{7} \right) \quad (1)$$

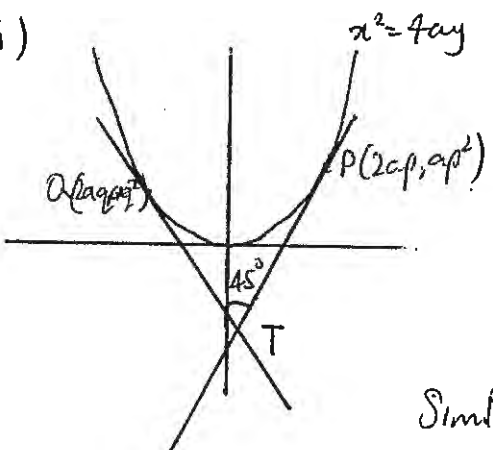
QUESTION:	Markers Comments
(d) $A = \pi r^2$ $\frac{dA}{dr} = 2\pi r$ ① $\frac{dr}{dt} = \frac{dr}{dA} \times \frac{dA}{dt}$ $= \frac{1}{2\pi r} \times 16$ $= \frac{16}{2\pi r}$ when $r=5$, $\frac{dr}{dt} = \frac{16}{10\pi}$ $= \frac{8}{5\pi}$ ① $= 0.51$ $\therefore$ radius is increasing at 0.51 m/min	*A lot of students forgot to find the answer correct ① to 2.dp.
(e) p = men who are colourblind = 0.08 q = men who aren't colourblind = 0.92	
$P(\text{exactly 6 men are colourblind}) = {}^{14}C_6 (0.08)^6 (0.92)^8$	①
(f) Area = $\int_0^{\pi/6} \sin^{-1} x \, dx$ = Area of rectangle - $\int_0^{\pi/6} \sin x \, dx$ ① $= \frac{\pi}{6} \times \frac{1}{2} - [-\cos x]_0^{\pi/6}$ $= \frac{\pi}{12} - (-\cos \frac{\pi}{6} - -\cos 0)$ ① $= \frac{\pi}{12} - (-\frac{\sqrt{3}}{2} + 1)$ $= \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1$ $= \frac{\pi + 6\sqrt{3} - 12}{12}$ ①	
*$(\frac{2}{\sqrt{3}} - 1)$ = 1 mark only *$(1 - \frac{\sqrt{3}}{2})$ = 2 marks only	*$\frac{\pi}{12} + \frac{\sqrt{3}}{2} = 2$ marks only

QUESTION: 12

Markers Comments

- (a) Let $\angle RAB = \alpha$
 $\angle RAB = \angle BCA$ (angle in the alternate segment)
 $\angle QCA = 180 - \alpha$ (straight angle)
 Let $\angle PAC = \beta$
 $\angle PAC = \angle ABC$ (angle in the alternate segment)
 $\angle PQB = 180 - \beta$ (co-interior angles are supplementary $QP \parallel AB$)
 $\therefore \angle PAC + \angle QPC$ are supplementary.
 $\therefore APQC$ is a cyclic quadrilateral.

Most students did this well

- (b) i)
- 
- $x^2 = 4ay$
 $4ay = x^2$
 $y = \frac{x^2}{4a}$
 $y' = \frac{x}{2a}$
 $m_1 = y'(P) = \frac{2ap}{2a} = p$
 Similarly, $m_2 = q$

$$\tan 45^\circ = \left| \frac{p-q}{1+pq} \right|$$

$$1 = \left| \frac{p-q}{1+pq} \right|$$

$$\therefore p-q = 1+pq$$

- ii) The tangent at P is $y = px - ap^2 \dots (1)$
 similarly at Q: $y = qx - aq^2 \dots (2)$

$$(1)-(2): (p-q)x - a(p^2 - q^2) = 0$$

$$x = \frac{a(p-q)(p+q)}{p-q} = a(p+q)$$

Most students were able to use angle between two lines.

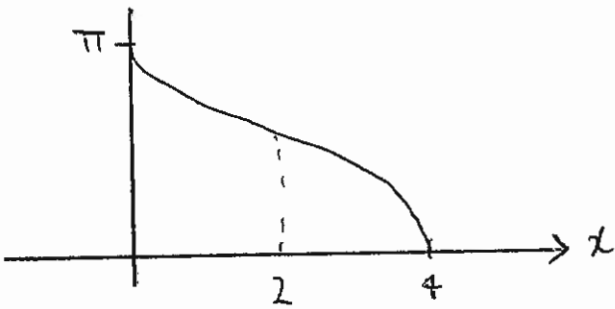
$$y = ap(p+q) - ap^2$$

$$= ap^2 + apq - ap^2$$

$$= apq$$

$$\therefore p+q = \frac{x}{a}, \quad pq = \frac{y}{a}$$

Use h & p for locust

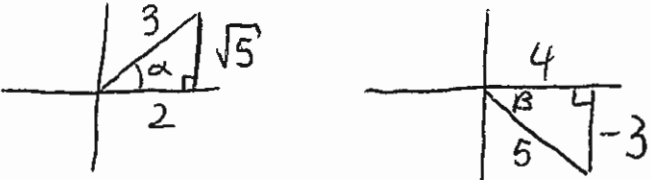
QUESTION: 12	Markers Comments
b) continued. From (i) : $(1+pq)^2 = (p-q)^2$ $= (p+q)^2 - 4pq$ Locus : $\left(1+\frac{y}{a}\right)^2 = \left(\frac{x}{a}\right)^2 - \frac{4y}{a}$ $\therefore x^2 + y^2 + 6ay = x^2$ (c) Domain : $-1 \leq \frac{x-2}{2} \leq 1$, Range $0 \leq y \leq \pi$. $-2 \leq x-2 \leq 2$ $0 \leq x \leq 4$  $\cos y = \frac{x-2}{2}$ $x-2 = 2\cos y$ $x = 2\cos y + 2$ $x^2 = (2\cos y + 2)^2$ $= 4\cos^2 y + 8\cos y + 4$ $V = \pi \int_a^b x^2 dy$ $= 4\pi \int_0^\pi (\cos^2 y + 2\cos y + 1) dy$ $= 4\pi \left[\frac{1}{2}y + \frac{1}{4}\sin 2y + 2\sin y + y \right]_0^\pi$ $= 4\pi \left[\frac{\pi}{2} + 0 + 0 + \pi \right] - 0$ $= 4\pi \left(\frac{3\pi}{2} \right)$ $= 6\pi^2 \text{ cubic units.}$	Only few students were able to find the equation of the locus. Students were able to get to this step. Some students could not $\int (\cos^2 y) dy$.

QUESTION: 12	Markers Comments
Consider $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$ <u>Multiply by x:</u> $x(1+x)^n = \binom{n}{0}x + \binom{n}{1}x^2 + \binom{n}{2}x^3 + \dots + \binom{n}{n}x^{n+1}$ <u>Differentiating:</u> $x(n(1+x)^{n-1}) + (1+x)^n = \binom{n}{0} + 2\binom{n}{1}x + 3\binom{n}{2}x^2 + \dots + (n+1)\binom{n}{n}x^n$ $(1+x)^{n-1}(xn + (1+x)) = \binom{n}{0} + 2\binom{n}{1}x + 3\binom{n}{2}x^2 + \dots + (n+1)\binom{n}{n}x^n$ $(1+x)^{n-1}(x(n+1) + 1) = \binom{n}{0} + 2\binom{n}{1}x + 3\binom{n}{2}x^2 + \dots + (n+1)\binom{n}{n}x^n$ <u>Substitute $x=1$:</u> $2^{n-1}(n+2) = \binom{n}{0} + 2\binom{n}{1} + 3\binom{n}{2} + \dots + (n+1)\binom{n}{n}$ <u>END OF QUESTION 12</u>	Student did not do this well.

Examination:

Level:

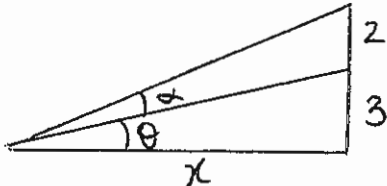
Year:

QUESTION: 13	Markers Comments
a) $\sin\left(\cos^{-1}\frac{2}{3} + \tan^{-1}\left(-\frac{3}{4}\right)\right)$ let $\alpha = \cos^{-1}\frac{2}{3}$ and $\beta = \tan^{-1}\left(-\frac{3}{4}\right)$  so $\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$ $= \frac{\sqrt{5}}{3} \times \frac{4}{5} + \frac{2}{3} \times \left(-\frac{3}{5}\right)$ 1 mark for negative sign. $= \frac{4\sqrt{5}}{15} - \frac{6}{15}$ $= \frac{4\sqrt{5} - 6}{15}$ 1 mark for final answer.	
b) $p(x) = ax^3 + bx^2 + cx + d$ i) but $a = 1$ since monic $p(0) = 0 + 0 + 0 + d$ $d = -6$ $\begin{array}{r} x^2 + 7 \overline{) x^3 + bx^2 + cx - 6} \\ \underline{-(x^3 + 0x^2 + 7x)} \\ 0 + bx^2 + (c-7)x - 6 \\ \underline{-(bx^2 + 0x + 7b)} \\ 0 + (c-7)x - 6 - 7b \end{array}$ $\begin{array}{lcl} c-7=1 & -6-7b=12 & \\ c=8 & -7b=18 & \\ & b=-\frac{18}{7} & \end{array}$ $\therefore p(x) = x^3 - \frac{18}{7}x^2 + 8x - 6$	1 mark for $a=1$ $d=-6$ 1 mark for division of polynomials. 1 mark for $c=8$ $b=-\frac{18}{7}$ Part b) was done poorly.

Examination:

Level:

Year:

QUESTION:	Markers Comments
b ii) $P(x) = x^3 - \frac{18}{7}x^2 + 8x - 6$ $P'(x) = 3x^2 - \frac{36}{7}x + 8$ $P(1) = \frac{3}{7}$ $P'(1) = \frac{41}{7}$ $x_2 = x_1 - \frac{P(1)}{P'(1)}$ $= 1 - \frac{\frac{3}{7}}{\frac{41}{7}}$ $= 1 - \frac{3}{41}$ $= \frac{38}{41} = 0.92$	Newton's method carry from previous error.
13c)  i) $\tan(\alpha + \theta) = \frac{5}{x}$ $\tan(\theta) = \frac{3}{x}$ $\theta + \alpha - \theta = \tan^{-1}\left(\frac{5}{x}\right) - \tan^{-1}\frac{3}{x}$ ii) $\frac{d\alpha}{dx} = \frac{1}{1 + \left(\frac{5}{x}\right)^2} \times -5x^{-2} - \frac{1}{1 + \left(\frac{3}{x}\right)^2} \times -3x^{-2}$ $= -\frac{\frac{5}{x^2}}{1 + \frac{25}{x^2}} + \frac{\frac{3}{x^2}}{1 + \frac{9}{x^2}}$ $= -\frac{\frac{5}{x^2}}{\left(\frac{x^2 + 25}{x^2}\right)} + \frac{\frac{3}{x^2}}{\left(\frac{x^2 + 9}{x^2}\right)}$	1 mark.

Examination:

Level:

Year:

QUESTION:	Markers Comments												
$\frac{d\alpha}{dx} = -\frac{5}{x^2+25} + \frac{3}{x^2+9}$ $\frac{d\alpha}{dx} = 0 \text{ for max}$ $\frac{5}{x^2+25} = \frac{3}{x^2+9}$ $5x^2+45 = 3x^2+75$ $2x^2 = 30$ $x^2 = 15$ $x = \pm\sqrt{15} \quad x > 0, \therefore x = \sqrt{15}$ <table><tr><td>x</td><td>3.5</td><td>$\sqrt{15}$</td><td>3.9</td></tr><tr><td>$\frac{d\alpha}{dx}$</td><td>0.01</td><td>0</td><td>-0.00195</td></tr><tr><td></td><td>+</td><td>0</td><td>-</td></tr></table> $\therefore x = \sqrt{15} \text{ is a max.}$	x	3.5	$\sqrt{15}$	3.9	$\frac{d\alpha}{dx}$	0.01	0	-0.00195		+	0	-	1 mark setting up $\frac{d\alpha}{dx}$ correctly. 1 mark for $x = \sqrt{15}$ 1 mark for testing $x = \sqrt{15}$ is max.
x	3.5	$\sqrt{15}$	3.9										
$\frac{d\alpha}{dx}$	0.01	0	-0.00195										
	+	0	-										
ii) $\alpha = \tan^{-1}\left(\frac{5}{x}\right) - \tan^{-1}\left(\frac{3}{x}\right)$ $= \tan^{-1}\left(\frac{5}{\sqrt{15}}\right) - \tan^{-1}\left(\frac{3}{\sqrt{15}}\right)$ $= 14^\circ.$	1 mark for answer.												
d) $\frac{\log_2 \frac{1}{2}x}{\log_2 \frac{1}{2}} \geq \log_2(4x-1)$ $\frac{\log_2 x^{-1}}{\log_2 2^{-1}} \geq \log_2(4x-1)$ $\frac{\log_2 x}{-1} \geq \log_2(4x-1)$ $x \geq 4x-1$ $3x \leq \frac{1}{3}$ But $x > \frac{1}{4}$ since $4x-1 > 0$ $\therefore \frac{1}{4} < x \leq \frac{1}{3}.$	1 mark change the base. 1 mark for $x \leq \frac{1}{3}$ 1 mark for testing $x > \frac{1}{4}$												

Examination:

HSC — Trial Examination

Level:

Extension 1 Mathematics

Year:

2018

QUESTION: 14	Markers Comments
a) $t = 4x^2 - 6x + 3$ (i) $\frac{dt}{dx} = 8x - 6$ $v = \frac{dx}{dt} = \frac{1}{8x-6}$ (ii) $a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ $= \frac{d}{dx} \left(\frac{1}{2} (8x-6)^{-2} \right)$ $= \frac{1}{2} \times -2 (8x-6)^{-3} \times 8$ $= \frac{-8}{(8x-6)^3}$ (iii) $t = 8 \text{ sec.}$ $4x^2 - 6x + 3 = 8$ $4x^2 - 6x - 5 = 0$ $x = \frac{6 \pm \sqrt{36 + 80}}{8}$ $= \frac{6 \pm \sqrt{116}}{8}$ $= \frac{3 \pm \sqrt{29}}{4}$ $x = \frac{3 + \sqrt{29}}{4}$ as $a > 0$ of the particle always moves to the right and velocity is never zero $v = \frac{1}{8 \left(\frac{3 + \sqrt{29}}{4} \right) - 6}$ $= \frac{1}{8 + 2\sqrt{29} - 6} = \frac{1}{2\sqrt{29}} \quad (\text{OR } 0.093 \text{ cm/s})$	well done Most of the students did it well very few wrote $(8x-6)^{-1}$ instead of $(8x-6)^{-3}$ good attempt - good to see students stating reasons for choosing positive value of x.

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b) (i) $T = R + Ae^{-kt}$ $\frac{dT}{dt} = -kAe^{-kt}$ $= -k(T-R)$ $\therefore T = R + Ae^{-kt}$ is a solution to the differential equation. (ii) $t = 0, T = 190^\circ\text{C}, t = 5, T = 85^\circ\text{C}$ $R = 25^\circ\text{C}$ $190 = 25 + Ae^0 \quad (e^0 = 1)$ $A = 165$, Also, $85 = 25 + 165e^{-5k}$ $\frac{60}{165} = e^{-5k}$ $\frac{4}{11} = e^{-5k}$ $-5k = \ln\left(\frac{4}{11}\right)$ $k = -\frac{1}{5} \ln\left(\frac{4}{11}\right)$ $63 = 25 + 165e^{-kt}$ $25 = 165e^{-kt}$ $\frac{38}{165} = e^{-kt}$ $t = -\frac{1}{k} \ln\left \frac{38}{165}\right$ $= 7.25 \dots \text{minutes}$ $= 7 \text{ min (to the nearest minute)}$	good attempt Most of the students found A, k correctly well done

Examination:

HSC - Trial Examination

Level:

Extension 1 - 1

Year:

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c) $P(\text{odd}) = p$, $P(\text{even}) = 1-p$ at most one even = Probability of (i) 0 even + prob of 1 $= {}^6C_6 p^6 (1-p)^0 + {}^6C_5 p^5 (1-p)$ $= p^6 + 6p^5 - 6p^6$ $= 6p^5 - 5p^6$ (ii) Product of scores is even, when it is at least one even. $1 - {}^6C_6 p^6 (1-p)^0$ $= 1 - p^6.$ d) $1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} \leq 2 - \frac{1}{n}$ <u>step 1</u>: $n=1$, LHS = 1 RHS = $2 - \frac{1}{1} = 1$ LHS = RHS, the result is true for $n=1$. <u>step 2</u>: let the result be true for $n=k$ $1 + \frac{1}{2^2} + \dots + \frac{1}{k^2} \leq 2 - \frac{1}{k}$ <u>step 3</u> To show that the result is true for $n=k+1$ i.e. $1 + \frac{1}{2^2} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2} \leq 2 - \frac{1}{k+1}$	good attempt. Most of the students could not do this part. well done for steps 1 and 2.

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LHS: $1 + \frac{1}{2^2} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2} \leq 2 - \frac{1}{k} + \frac{1}{(k+1)^2}$ (using the assumption) In order to show that $1 + \frac{1}{2^2} + \dots + \frac{1}{(k+1)^2} \leq 2 - \frac{1}{k+1}$ enough to show that $2 - \frac{1}{k} + \frac{1}{(k+1)^2} \leq 2 - \frac{1}{k+1}$ i.e. we $\frac{1}{k+1} - \frac{1}{k} + \frac{1}{(k+1)^2} \leq 0$ $\frac{k(k+1) - (k+1)^2 + k}{(k+1)^2} \leq 0$ i.e. $\frac{\cancel{k^2} + k - k^2 - 2k - 1 + \cancel{k}}{(k+1)^2} \leq 0$ i.e. $\frac{-1}{(k+1)^2} \leq 0$ which is true. $\therefore$ using the Principle of Mathematical Induction, the property is true for all $n \geq 1$	Most of the students were lost in proving the inequality step 3